

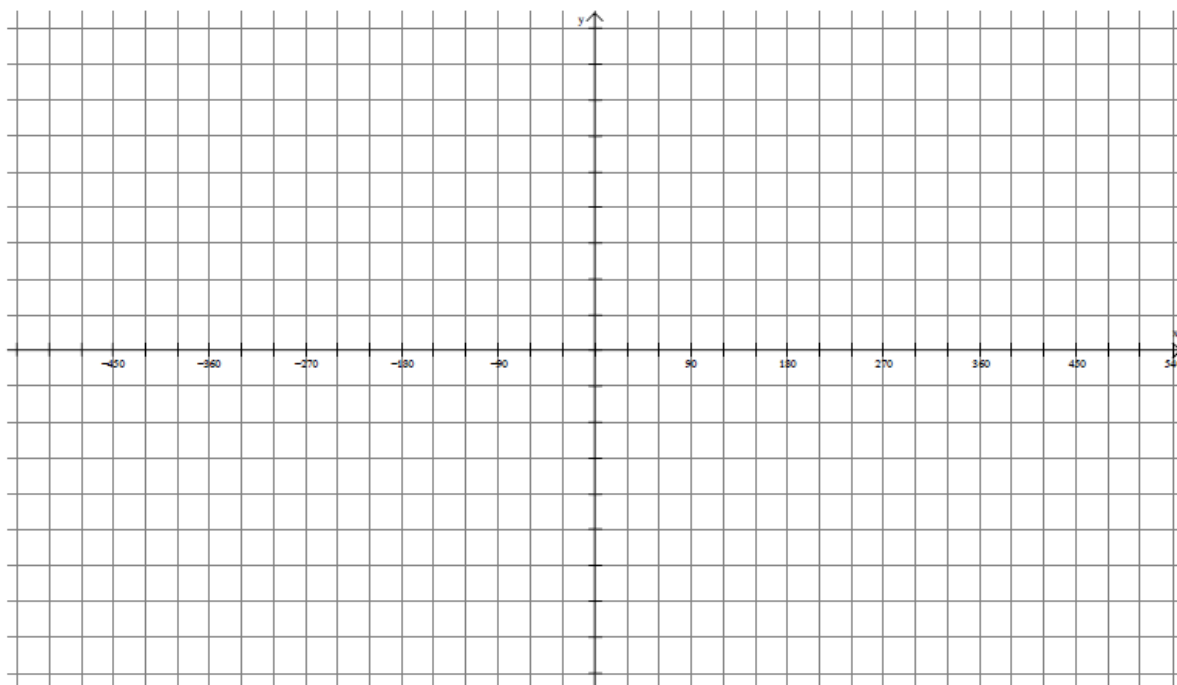
5.3 Transformations of Sine and Cosine Worksheet #1

MCR3U

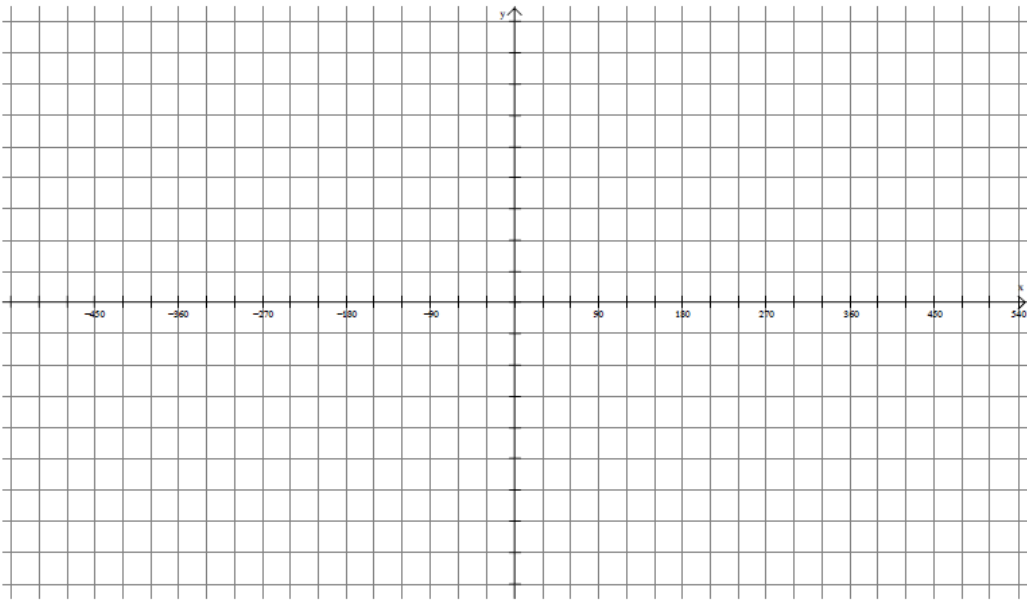
Instructions: For each of the following start with two cycles of the parent graph (-360 to 360). Transform key points to produce the new graph.

1) Using transformations, graph two cycles of the following trigonometric functions. State the period, phase shift, amplitude, and vertical displacement.

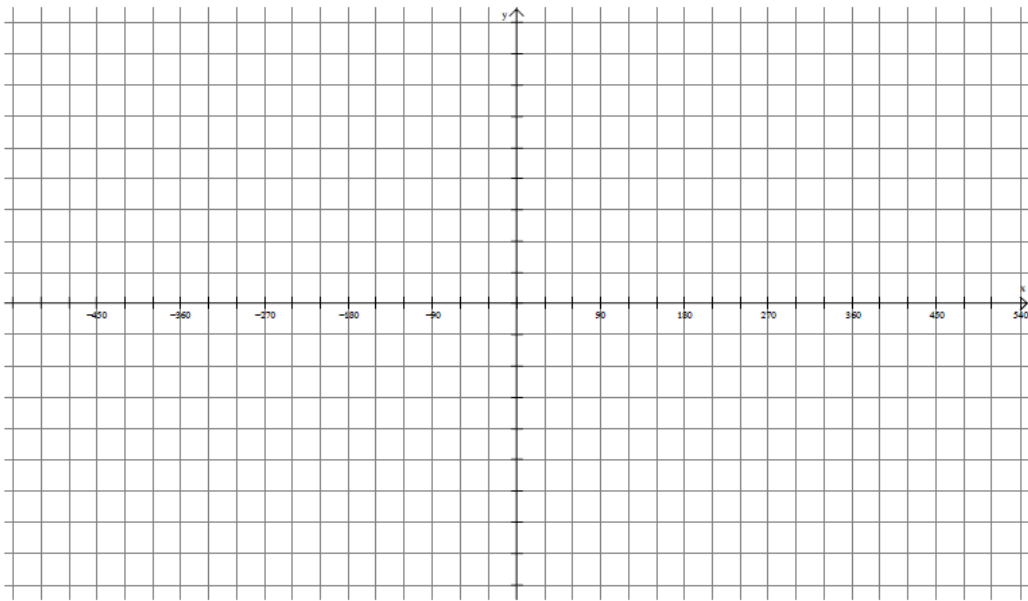
a) $y = 4\cos(\theta + 180^\circ) + 2$



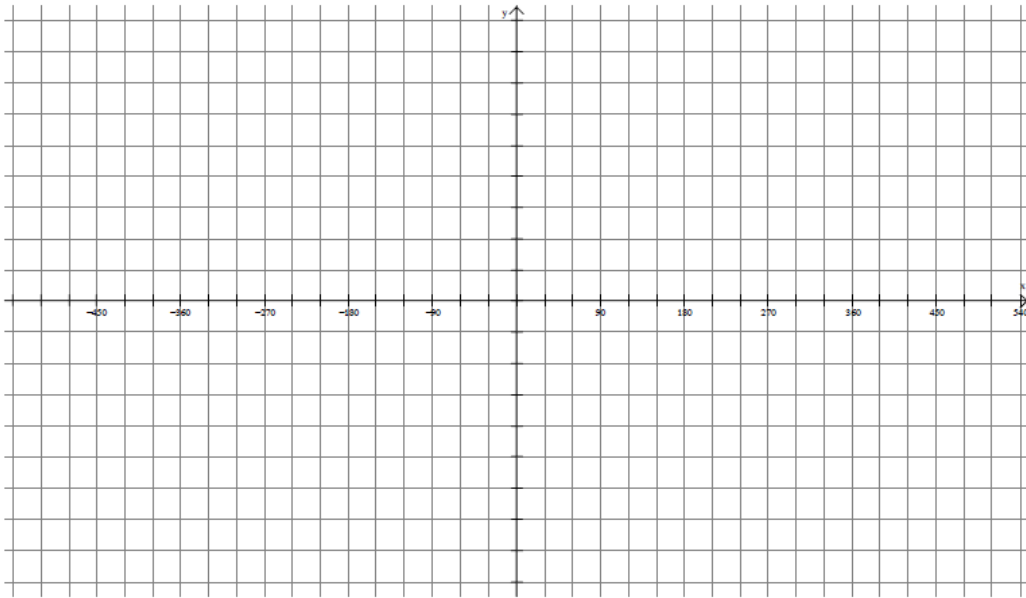
b) $y = \sin\left[2\left(\theta + 30^0\right)\right] + 3$



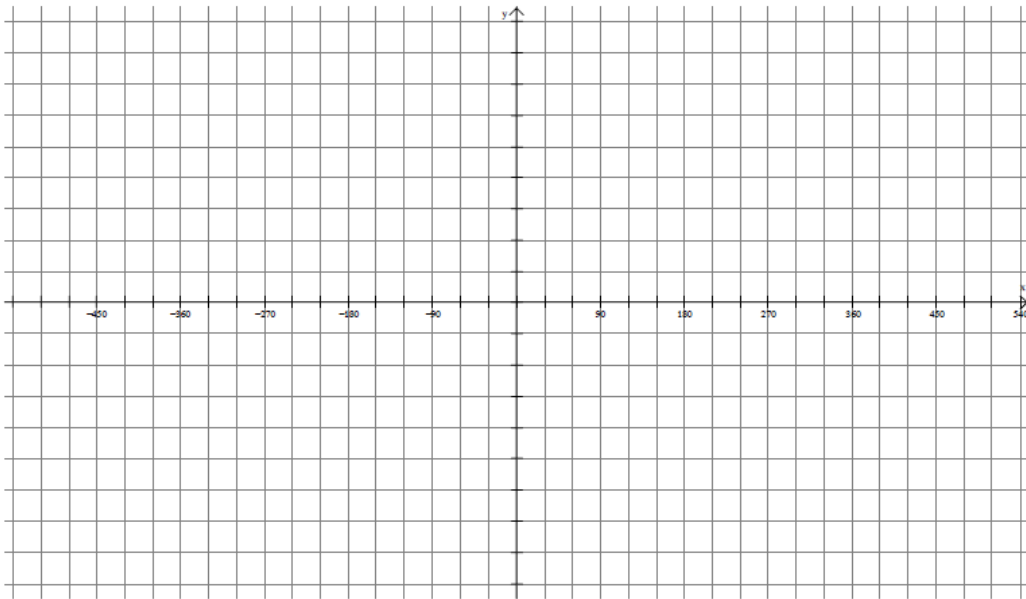
c) $y = 2\sin\left[3\left(\theta - 180^0\right)\right]$



d) $y = -3\cos(2\theta) + 4$



e) $y = \frac{1}{2}\sin(\theta + 45^\circ) - 3$



f) $y = 3\cos\left[2\left(\theta - 60^\circ\right)\right] + 4$

